

CPAP Device Estimation Tool v1.1

What the tool does

This tool estimates how many CPAP (Continuous Positive Airway Pressure) breathing-support devices a hospital newborn unit needs. It takes two simple inputs - average monthly admissions and the hospital's level of care - and returns a device count with a 95% confidence range. It was built from patient-level records (2021-2026) across 105 hospitals in five African countries (Ethiopia, Kenya, Malawi, Nigeria, Tanzania), in partnership with the NEST360 Alliance.

Capacity to provide quality CPAP (staffing, infrastructure, oxygen supply, monitoring, etc.) must be assessed at each hospital before this estimate is used to inform procurement decisions.

How it works

Base devices = ROUNDUP(Admissions × Eligibility rate × Length of treatment / 30 × Surge factor)

Maintenance buffer = 0 if Base devices = 0; otherwise at least 1, rounded up from Base devices × Maintenance rate

Devices needed = Base devices + Maintenance buffer

In plain terms: how many babies need CPAP each month, multiplied by how long each one typically stays on a device, gives the average number of devices in use at once. We then add a margin for busier-than-average periods (surge), and separately add a maintenance buffer for devices temporarily out of service for repair - calculated on its own so it doesn't disappear to zero for smaller hospitals, as it would if folded into one combined rounding step.

Key methodological decisions

- Eligibility (who needs CPAP) is based on a hybrid of clinical criteria, observed rates at well-supplied hospitals, and expert opinion - and a baby who received CPAP always counts as having needed it, regardless of documentation.
- Length of treatment uses a competing-risks statistical method (Restricted Mean Time Lost, RMTL) that separates babies who died on CPAP from babies who recovered, so that early deaths do not distort the treatment-duration figure.
- Hospitals are grouped into three levels aligned with WHO newborn care standards: Level II Basic (small district/mission hospitals), Level II Comprehensive (larger district/regional hospitals), and Level III (tertiary/referral hospitals).
- Surge (2.0×) and maintenance buffer (3.4%) are fixed values, from expert judgment and equipment audit data respectively, applied the same way across all hospital levels.
- Devices needed are calculated in two rounded steps - demand and surge together, then a maintenance buffer rounded separately and always at least one device once there is real demand - so the buffer doesn't vanish for smaller hospitals, without the extra inflation that would come from rounding every step independently.

What we found

Level of care	Eligibility	Treatment length	Surge	Maintenance
Level II Basic	18.2%	4.24 days	2.0×	3.4%
Level II Comprehensive	16.2%	4.25 days	2.0×	3.4%
Level III	28.9%	4.28 days	2.0×	3.4%

All figures are empirically derived from data, with confidence intervals available in the underlying analysis.

Strengths

- Parameters are derived from historical hospital data.
- Accounts for hospitals where CPAP supply is constrained or coverage data is unreliable, rather than treating all hospitals as equally trustworthy sources.
- Separates death from recovery when estimating treatment duration, avoiding bias from competing risks (death) before successful weaning or undocumented wean time for babies that survive.
- Every estimate comes with a 95% confidence range, making the underlying uncertainty explicit.
- Aligned with WHO's newborn care level-of-care framework (WHO, 2020), so results are relatable across health systems in various settings.

Caveats and limitations

- Reflects current practice, not a clinical guideline for ideal treatment length: the tool answers “how many devices are needed given how hospitals currently operate,” not “how many would be needed under idealized care.”
- Documentation quality can shift the numbers in either direction; we've corrected for one known issue (missing treatment end-dates) but cannot rule out others (e.g. recording delays or wrong moderately inaccurate time inputs).
- Length of treatment is blended using each hospital level's typical mix of patients by birth weight, and not any single hospital's exact case mix, which may not fit hospitals with an unusual patient profile/case mix.
- The Level III estimate is based on fewer hospitals than the two Level II categories, so its confidence range is wider and less precise.
- This tool has not yet been validated against independently measured device counts or usage in real hospitals.
- Level-of-care labels follow the WHO classification framework for consistency across countries; actual on-the-ground capacity at a given hospital may differ and should be verified directly, not assumed from the label alone.
- Device cleaning, disinfection, and preparation time between patients is not yet factored into these estimates, and would increase the effective number of devices needed beyond what is shown here.

Output and access

The tool is available as a simple, free web calculator, part of NEST360's Device Quantity Estimation Tools initiative: a user enters monthly admissions and selects a hospital level, and receives a device estimate with its confidence range and a breakdown of the calculation. This document is also available as a downloadable PDF directly from the tool page.

Web link: <https://cpap-calculator.zahara-quantum.com/cpap.html>

Next steps

- Planned: independent validation against measured device use in real hospitals (in and outside NEST360 network).
- Planned: extending the same approach to other neonatal equipment beyond CPAP.
- Under discussion: methods for rolling up hospital-level estimates into a national device-need total.

Questions about this tool or its methodology are welcome at any time - please reach out to the project team (danicakumara@gmail.com, jwainaina@rice.edu) or the NEST360 Alliance (info@nest360.org)